

Yazdan Rasoulzadeh

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EDUCATION

University of Waterloo & Wilfrid Laurier University

Waterloo, ON

Bachelor of Computer Science & Business Administration (Double Degree)

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, Rust, Java

Frameworks/Tools: Docker, Git, MySQL, SQLite, PostgreSQL, GitHub Actions

Other: Linux, REST APIs, Azure, AWS

PROJECTS

Calcite AI Agent | Python, Docker, Rasa, GitHub Actions

July 2025

- Engineered a modular **NLP backend** using **Rasa**, eliminating **5+** hours of manual bookkeeping weekly.
- Containerized the desktop application with **Docker**, supporting consistent local cross-platform execution.
- Created a **CI/CD pipeline** leveraging **Github Actions**, cutting testing and deployment times by **over 80%**.

RustAuth Microservice | Rust, Axum, PostgreSQL, JWT, tower-http

June 2025

- Developed a robust authentication **microservice** implementing user registration, login, and JWT-based session management, streamlining user access.
- Engineered **REST API** endpoints with Axum and **PostgreSQL** for secure credential management via hashed storage and JWT validation **middleware**, enhancing security.
- Optimized performance to handle **6,600 requests/sec** with **0.5 ms median latency** and **99% completion under 8.2 ms**, ensuring rapid, reliable authentication.

Scratch HTTPS Server | C, OpenSSL, Sockets

April 2025

- Built a **secure HTTPS server** in C from scratch using **OpenSSL** and low-level **socket programming**, establishing encrypted communication channels.
- Implemented robust handling for GET and POST requests with dynamic parsing and accurate status codes, ensuring comprehensive HTTP protocol compliance.
- Supported multiple **concurrent** users through **multithreaded** request handling and safe memory management, processing **217 requests/sec** with **50 ms median latency** for **1000+ users**.

EXPERIENCE

Director of Finance

Jan 2025 – Apr 2025

UW Mathematics Society

Waterloo, ON

- Ensured accurate and timely fund distribution for **9000+** students by processing 30+ reimbursement requests using **standardized** digital intake forms and structured spreadsheet workflows.
- Improved financial tracking accuracy by reconciling weekly cash deposits through internal logs and **dual-verification**, reducing the risk of manual errors.
- Simplified audit preparation by maintaining well-organized, consistently **formatted** financial records using spreadsheet systems.

Hackathon Theme Winner

Feb 2025

MakeUoft

Toronto, ON

- Competed among **250+** participants in Canada's largest make-a-thon, constructing an end-to-end IoT solution, placing **1st** in the environment theme.
- Engineered core **backend** logic in C++, Arduino, Python, enabling seamless hardware-to-cloud communication.
- Enhanced contextual decision-making, by leveraging the **Gemini API**, enhancing system intelligence